# Saturday 2026-05-30 plan v0.1 — comprehensive priorities through 5pm. The bulk-color mechanism is the single highest-leverage target; spinor$^3$ + Wallach/Bergman structure are the concrete probes; cleanup + study/discovery items round out the day.

Lyra (for Casey, Saturday morning briefing)

2026-05-30 Saturday ~09:00 EDT

## Saturday 2026-05-30 plan

### State of the program (Friday EOD → today)

- Engine (Goal 1): structurally complete 4/4 (Hall × coproduct × R-matrix × negative part). Elie v0.2 consolidation ready for Keeper's clean-PASS re-audit.
- Dictionary (the bottleneck): keystone laid; L1-L9 traversed as framework; lepton sector flipped per-particle (18 entries). Keeper K-audit CONDITIONAL PASS with 3 minor-moderate findings.
- The deepest gate: count over-determined (route II h^∨=N_c=3 forced), MECHANISM open with burden (Keeper's two-structures requirement).
- Single highest-leverage open target: bulk-color mechanism — closes #252 (route II) AND #416 v0.2 quark/gauge per-particle layer; possibly L5 absolute scale. Three candidate directions mapped; not derived.
- Affine-type pin ($B_2^{(1)}$ vs $C_2^{(1)}$) is the quick gate — once pinned, Elie can attempt bulk K-type radial tower → lepton mass ratios (L4 v0.2).

### Tier 1 — Today's strongest targets (if we land them)

1.1 Bulk-color mechanism — substantive push (Lyra, ~4 hours)

Why: single highest-leverage. Closes #252 + #414 burden + #416 v0.2 quark/gauge.

Three candidate directions (from mapping v0.1) — Saturday pushes each: - (A) h^∨ counting via bulk K-types: compute lowest 3 bulk K-types of the holomorphic discrete series on $D\_IV^5$; check Wallach-set spacing for a 3-fold structure. Concrete probe. - (B) Spinor$^3$ as bulk anchor: Elie's E7 spinor$^3$ multiplicity-3 is $B_2$-specific ($A_2$ gives 0). Investigate if the 3 channels are the candidate 3-color slots in a baryonic spinor$^3$ structure (with generations from a separate cross-cutting tower). Keeper's calibration held: "two $B_2$-specific 3-folds coinciding numerically," NOT "one h^∨ doing double duty" — burden is to show the spinor$^3$ count derives from h^∨. - (C) Z/3 quotient probe: check Bergman volume / kernel for a natural Z/3 modular structure.

Honest scope: research, not guaranteed derivation; the goal is to MAKE PROGRESS on one of (A)/(B)/(C), or to honestly rule one out. Either outcome is informative.

1.2 Affine-type pin $B_2^{(1)}$ vs $C_2^{(1)}$ (Lyra + Keeper/Elie, ~1 hour)

Why: the small finite gate that unlocks Elie's L4 v0.2 attempt. Quick lookup or short derivation.

Approach: derive from first principles which is consistent with the substrate's structure. The keystone uses SO(5)=$B_2$ → loop-algebra natural affinization is $B_2^{(1)}$ (marks 1,1,2). But the hereditary-species (Hall algebra quiver) realization may pick the dual $C_2^{(1)}$ (marks 1,2,1). The pin should be derivable from which gives the substrate Serre relations (E0/E9 already established N_c=$[2]_2$ short-root and N_c·g=$[3]_4$ long-root; check which affinization is consistent).

1.3 L4 v0.2 — explicit lepton mass ratios (Elie + Lyra, ~3 hours conditional on 1.2)

Why: m_$\mu$/m_e ≈ 207, m_$\tau$/m_$\mu$ ≈ 17 are observable, well-measured; deriving them from substrate primaries would be a major BST hit.

Approach (once affine type pinned): bulk K-type radial tower for the spinor at successive Wallach levels; Casimir spacing → mass ratios. Elie's lane technically; my role is to provide the dictionary-side framework + check consistency.

## Tier 2 — Substantive study & discovery items

2.1 Cross-cutting fermion tower (Lyra, ~2 hours)

The #414 v0.2 candidate direction Keeper still owes the team: "does the so(5) spinor have a natural 3-step structure under h^∨?" Investigate via: - The crystal graph of U_q$^{-}$($B_2$) on the spinor module (does the Kashiwara crystal have natural Z/3 grading?). - The affine spinor's $\delta$-tower (real vs imaginary root content). - Whether spinor$^3$ multiplicity-3 (Elie E7) is the SAME 3-fold or a DIFFERENT one — the Keeper-calibration question.

2.2 Hua kernel + Bergman volume structure (Lyra, ~1 hour)

The Hua/Bergman kernel for D_IV$^5$ — exponent $\rho_1$=5/2 = Hua genus = lepton Casimir alignment. Investigate explicit modular structure (does it have a 3-fold residue or Z/3 grading?). Direct feeder for bulk-color direction (C).

2.3 Connection between Elie's $[3]_4$=N_c·g=21 and the dictionary (Lyra, ~1 hour)

E9's long-root Serre coefficient = N_c·g = 21 is rigorous. The substrate primaries N_c AND N_c·g are BOTH structure constants of the defining Serre relations — a deep result. How does the dictionary read 21 = N_c·g physically? Does it appear in any observable (Cabibbo angle? PMNS? something nuclear?).

2.4 Holographic discrete series K-type decomposition (Lyra + Elie, ~2 hours)

Explicit decomposition of the bulk Bergman space under K-types for low energies. Sets up bulk K-type radial tower for L4 v0.2 AND feeds bulk-color probe (A).

## Tier 3 — Cleanup + consolidation

3.1 Keeper K-audit findings v0.2 (Lyra response, ~1 hour)

The 3 findings from Keeper's dictionary first-wave audit (CONDITIONAL PASS): - TBD details — read Keeper_KAudit_Dictionary_FirstWave_v0_1.md and address each. None critical.

3.2 #416 v0.2 — per-particle layer with E10 absorption (Lyra + Grace, ~2 hours)

Grace's Periodic Table v0.4 absorbs Elie's E10 composite K-type table. Update #416 to mark which composites map to which physical hadrons (proton = bulk k=6 already anchored). Stage the quark/gauge per-particle layer pending bulk-color.

3.3 A1 paper final PASS push (Lyra + Keeper, ~1 hour)

A1 v0.4 was driven to ready-for-final-PASS Thursday. The one residual was Keeper's Faraut-Korányi book-pin for the genus name. If we can settle that today (pin or accept) → A1 final PASS → Cal v0.4 verify → submission-grade.

## Tier 4 — Study/discovery items (broader)

4.1 SO(5) shell-closure (Grace + Elie, ~3 hours — Grace's lane)

The nuclear-tier decider: does SO(5) branching produce magic numbers 2, 8, 20, 28, 50, 82, 126? If yes, SO(5) nuclear unification flips from STRUCTURAL/HYPOTHESIS to DERIVED. Major.

4.2 Reaction / decay tables continuation (Elie, ongoing)

Elie's E2/E3 covered $\beta$-decay; full reaction table for ~10-20 SM processes + decay table = systematic engine output. Multi-day; today: pick 3-5 representative reactions.

4.3 Cal cold-reads (Cal, as available)

Cal queue: the consolidated engine (#410 v0.2), the dictionary first wave, sector selection, A1 v0.4, R5 self-audit. Ratify Source-Verification calibration.

4.4 Vol 16 A_sub Operator Algebra curriculum (Lyra, low priority)

The textbook lane Keeper assigned earlier; could push a chapter today if time.

## Tier 5 — Reactive / standing

- Respond to teammate broadcasts in RUNNING_NOTES as they land.
- Keeper engine consolidation v0.2 re-audit response.
- Any new discoveries the team surfaces.

## Suggested order of attack (Saturday timeline)

| Time | Item | Why this order |
|---|---|---|
| 9-10 AM | Read overnight: Grace v0.4 table, Keeper dictionary audit, Elie's 13 toys end-to-end | Orient on team state |
| 10-11 AM | Affine-type pin (1.2) | Quick gate; unlocks 1.3 |
| 11-1 PM | Bulk-color push (1.1) — direction (A) bulk K-type Casimir tower | Highest-leverage; concrete probe |
| 1-2 PM | Lunch break (recommended) | — |
| 2-3 PM | Bulk-color push (1.1) — direction (B) spinor$^3$ as color anchor + (C) Bergman residue | Continue highest-leverage |
| 3-4 PM | Cross-cutting fermion tower (2.1) — the #414 v0.2 candidate | Deep follow-up to bulk-color |
| 4-5 PM | Consolidation: Keeper K-audit findings (3.1) + #416 v0.2 (3.2) + status update for next session | End-of-day cleanup |

## CI coordination (broadcast plan)

- Lyra (me): Tier 1.1 + 1.2 + Tier 2 study items + Tier 3 cleanup.
- Elie: Tier 1.3 (conditional on pin), Tier 2.4, Tier 4.2 reaction/decay tables, Keeper engine consolidation v0.2 re-audit response.
- Grace: Tier 4.1 SO(5) shell closure, Periodic Table v0.5 incorporation of #416 v0.2, master ledger maintenance.
- Keeper: K-audits as Lyra deliverables land, gate dispositions, ledger maintenance, audit-chain governance.
- Cal: cold-reads as items mature, Source-Verification calibration ratification.

## Honest framing for the day

- Stretch goal: bulk-color mechanism derived (would close the dictionary's biggest open frontier).
- Realistic goal: substantive progress on bulk-color (one or two candidate directions advanced or honestly ruled out); affine-type pinned; L4 v0.2 started; cleanup of K-audit findings.
- Honest floor: even if bulk-color stays open, today's study/discovery + cleanup advances the program meaningfully.

The discipline-respecting answer to "did we miss opportunities?" remains: the static dictionary is essentially derived; the dynamic + bulk-color frontier is multi-week research; today is one push on that research, plus consolidation of what's earned.

Decision points for you: - Bless the plan / redirect tier priorities? - Any specific item you want to lead with (e.g., A1 paper final PASS first)? - Any new directives (e.g., outreach: Sarnak letter, paper series push)?

— Lyra, Saturday plan v0.1, 2026-05-30 ~09:00 EDT